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Applications of artificial intelligence for membrane separation: A review

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ABSTRACT

The incorporation of Artificial Intelligence (AI) techniques into membrane-based systems has transformed their design, optimization, and management in various fields, notably, Reverse Osmosis (RO) and Ultrafiltration (UF). This comprehensive review evaluates the strategic application of AI methodologies across different membrane separation processes, furnishing practical insights for methodology selection and refinement. Ranging from predictive Machine Learning (ML) models for microfiltration to sophisticated neural networks for fouling alleviation in ultrafiltration, this review elucidates the alignment of AI approaches with specific application requisites. Despite persistent challenges such as data scarcity, interpretability of models, and computational demands, promising frontiers are emerging, including the amalgamation of AI with sensor technologies for real-time monitoring and control, and the utilization of generative adversarial networks for membrane material innovation. As the domain of AI in membrane separation progresses, interdisciplinary collaboration becomes paramount in surmounting existing obstacles and exploiting nascent prospects. This review underscores the necessity of comprehending AI methodologies tailored to diverse membrane separation context, steering future investigations towards heightened efficacy, sustainability, and ingenuity.

1. Introduction

Membranes have been widely used since 1960s as selective barriers between phases [1–3], finding applications in various industries such as water and wastewater treatment [3], desalination [4], gas separation [5], resource recovery [6,7], and food and beverage [8]. Membrane processes offer high separation capability, low energy consumption, and a small environmental footprint [9]. Given the challenges of water scarcity and pollution [10], along with the need to reduce greenhouse gas emissions [11], membranes play a crucial role in ensuring the availability of fresh water, capturing and purifying carbon dioxide and methane, as well as recovering valuable resources and nutrients [12]. Additionally, membranes are increasingly important in the food and beverage sector, enabling improved filtration, emulsion, and separation capabilities for particles of different sizes [13].

In order to unlock the full potential of membrane processes, significant efforts in membrane fabrication and process parameters optimization have been carried out to increase separation capacity, and membrane lifetime and to reduce energy consumption [14]. A range of mechanistic mathematical and empirical methods have been implemented, leading to considerable progress in this field. Nevertheless, these methods could not yield sufficient satisfaction due to the increasing complexity of systems and the need for more accurate modeling and predictions [15]. As an alternative, in recent years, hybrid models, namely Artificial Intelligence (AI)-based hybrid models, have arisen from advances in both fields of phenomenological and empirical modeling. With this approach, a more complex and trustworthy model can be developed while reducing the computational load [16].

AI is focused on the development of smart systems which tackle problems in a comparable way to human intelligence systems [1]. AI is already vastly applied in different fields, such as smart cities, big data, and healthcare, focusing on performance prediction, system efficiency, and AI integration with the process [14]. The revolutionary impact of AI has been so progressive that it has been referred to as the "4th paradigm of science" and the "4th industrial revolution" [17]. Machine learning, as a subfield of artificial intelligence, has experienced fast progress in recent years. Statistical algorithms are at the core of machine-learning applications, whose performance improves with training [18]. This type of approach is particularly useful when dealing with complex problems involving large combinatorial spaces, or nonlinear processes, which are either impossible to solve by traditional methods or can only be handled at a great computational cost. More recently, Deep Learning (DL) as a subclass of machine learning has gained huge attention. In DL, multiple layers are utilized to progressively obtain higher-level features from the raw data input [19]. Subject to variability due to data quality

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Abbreviations

AdaBoost	Adaptive Boosting
AI	Artificial Intelligence
ANFIS	Adaptive Neuro-Fuzzy Inference System
ANN	Artificial Neural Network
BAT	Bagged Trees
BLR	Bayesian Logistic Regression
BP(*A*)NN	Back Propagation (Artificial) Neural Network
CATBoost	Categorical Boosting
CFMF	Crossflow Microfiltration
CNN	Convolutional Neural Network
CP	Concentration Polarization
DL	Deep Learning
DNN	Deep Neural Network
DS	Draw Solution
DT	Decision Trees
ED	Electrodialysis
ERT	Extremely Randomized Trees
FIE	Flux Improvement Efficiency
FO	Forward Osmosis
FS	Feed Solution
GA	Genetic Algorithm
GBDT	Gradient Boosting Decision Tree
GBR	Gradient Boosting Regressor
GP	Gaussian Process
GPRM	Gaussian Process Regression Model
GUI	Graphical User Interface
KNN	K-Nearest Neighbours Algorithm
LR	Linear Regression
LSTM	Long Term Short Term Memory
MAE	Mean Absolute Error
MART	Multiple Additive Regression Trees
MBR	Membrane Bioreactor
MD	Membrane Distillation
MDP	Markov Decision Process
MF	Microfiltration
ML	Machine Learning
MLP	Multilayer Perception
MLR	Multiple Linear Regression
MPNN	Message-Passing Neural Networks
MSE	Mean Square Error
NF	Nano Filtration
OCT	Optical Coherence Tomography
OSN	Organic Solvent Nanofiltration
PLS	Partial Least-Squares
PSO	Particle Swarm Optimization
QSAR	Quantitative Structure-Activity Relationship
RF	Random Forest
RhB	Rhodamine B
RMSE	Root Mean Square Error
RO	Reverse Osmosis
RSF	Reversal Solute Flux
RSM	Response Surface Methodology
SL	Supervised Learning
SMILES	Simplified Molecular-Input Line-Entry System
SVM	Support Vector Machine
SVR	Support Vector Regressor
TDS	Total Dissolved Solids
TMP	Trans Membrane Pressure
TOC	Total Organic Carbon
TSS	Total Suspended Solids
UF	Ultra Filtration
XGBoost	Extreme Gradient Boosting

Symbols

ω	Direction vector
C	Adjustment factor
b	Intercept of the regression function
ε	Non-sensitivity coefficient
ξ_i	Slack variable
ξ_i^*	Slack variable
$\phi(x_i)$	Higher-dimensional hyperspace for the input vector x_i.
n	Number of data points
P_i	Predicted output of the model
$\overline{y_i}$	Sample data mean
y_i	Actual output
R^2	Coefficient of The Determination

and availability, the prediction of chemical composition, structure, and properties have been enhanced by DL in several cases [18].

This review provides a detailed bibliometric analysis of current literature on applying artificial intelligence (AI) in developing and optimizing membrane separation technologies. It aims to help researchers identify emerging trends and breakthroughs in this innovative field, where AI has shown significant potential for advancing membrane technology. Initially, the effort has been to discuss the general idea behind AI and to introduce the relevant applied algorithms in membrane research. Then, a summary of up-to-date studies has been given considering the type of membrane and application intended to obtain a categorical view of the subject. Finally, considering the current status and technological barriers with membranes, the potential direction of AI-assisted research of membranes is given.

2. Methodology

2.1. Data acquisition and screening

The bibliographic data of membrane research from 1 January 2013 to 15 March 2023 were retrieved from the Web of Science database. The present study is limited to only research articles obtained from SCOPUS and Web of Science and screened following the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) [20] step-by-step guidelines. First, 409 journal articles were obtained from the SCOPUS database with relevant keywords. The data selected for each publication includes title, author, affiliation, country, (authors-defined) keywords, publication year, and citation count. In the pre-screening process, review articles and articles without access to the full text were omitted. Of the remaining 313 articles, 109 of them were discarded due to not falling into the membrane filtration category.

2.2. Data pre-processing

Before performing the analysis, the retrieved data needed to be pre-processed to account for the keywords being mentioned by authors in different forms or with synonym terms. An example of this is Artificial Neural Networks (ANN). These terms and words are being combined to give an accurate counting of the frequencies they have appeared in research articles. Table S1, in the Supplementary Data, tabulated the list of similar keywords along with the replacement terms which is the most repeated keyword compared to its similar terms.

2.3. Data analysis

In order to visualize the evolution and trend of research topics during

the past decade, the pre-processed data is imported and analyzed by MS Excel and VOSviewer. Countries involved, year of publication, publishing journals, and co-occurrence of keywords are some of the indications analyzed that shed light on the research hot spots by connecting the dots in published journals.

3. Artificial intelligence in membrane research

3.1. Spatio-temporal evolution of AI-assisted membrane research

The spatiotemporal trend of published papers since 2013 on AI-assisted membrane research reveals particularly interesting information. Fig. 1a demonstrates a yearly trend of published articles by country in which almost steadily few papers have been published until the year 2018 with a maximum of 26 in 2016. This follows an exponential growth in publications where the number of published papers has increased drastically with a maximum of 122 papers in 2022. It is also observed that the number of contributing countries has increased to 49 with several Middle Eastern and African countries among them as shown in Fig. 1 while several countries being the field leaders including China, the United States of America, South Korea, and India. The reason for this increasing interest among these countries can be due to the increased efforts in these regions to address their local challenges using membrane technologies, such as industrial innovation, clean water, and healthcare. Other reasons for the substantially increasing number of papers can be found in the technological advancement of both membrane science and artificial intelligence, as well as increased awareness of the importance of this field and funding dedicated to it and the expansion of cross-disciplinary research where researchers from both fields collaborate to solve problems that require expertise from both fields.

3.2. Interconnections of research topics with co-occurrence of keywords

The interconnection of pre-processed keywords used in AI-assisted membrane research since 2013 is illustrated in Fig. 1b. These keywords have been used at least three times in different sources. Three major focus areas can be detected from this network: 1) Different membrane processes, such as RO, membrane bioreactor (MBR), and UF. 2) Process-related terms namely such as fouling, an inevitable operating challenge accounting for performance decline and reduced lifespan of membrane systems. Accurate detection and quantification of fouling have remained a challenging subject for researchers [14] and this can explain the reason for the extensive application of AI in fouling modeling. 3) AI terms are the 3rd main category of keywords. ML, AI itself, and ANN in particular which is at the center of attention by being the most repeated keyword. In the following section, it will be disclosed that the reason for this repetition is essentially the popularity of this AI technique among membrane researchers [21]. On the other hand, there is a shift in the research trend with more focus on emerging subjects, such as micropollutant removal [22], and coupling membrane systems with renewable energy resources for energy management and achieving the circular economy goal.

3.3. Progress in artificial intelligence models and algorithms for membrane research

Mechanistic modeling and non-mechanistic modeling are two distinct approaches used in studying complex systems, including membrane processes. Mechanistic models are based on fundamental physical, chemical, or biological principles and aim to explain system behavior using established laws and theories. These models provide insights into underlying mechanisms but can be limited by an incomplete understanding of physicochemical processes and may not generalize well to new conditions. In contrast, non-mechanistic or data-driven models focus on identifying patterns and relationships within data without relying on predetermined theories. These models are often more flexible and better suited for handling large, complex datasets, but may lack explanatory power regarding underlying mechanisms. Conventionally, various membrane types and membrane processes have been studied using traditional mechanistic models. These models and modeling systems serve a function that cannot be replaced, but in some instances, our understanding of the physicochemical processes and interactions is not sufficient to develop an effective and accurate model [23]. Additionally, computational tools are essential when dealing with large and complex amounts of data resulting from monitoring membrane processes, allowing meaningful information to be extracted. The goal of mechanistic modeling is to explain the system and the experimental data in accordance with a priori understanding, while the goal of non-mechanistic modeling is to determine the system phenomena and dynamics that are inherent in the data. Moreover, the mechanistic models have a specific meaning that they have been developed based on data collected from a single experiment under certain operating conditions, which can mislead the model for new data. Complicated equation series formed because of the high number of variables in mechanistic models is another problem with these models which leads to complications of the model and a drastic increase in computation time [24].

AI-based technology has been propounded as an efficient tool for predicting membrane separation and filtration processes that can be used in the future, in order to overcome the limitations of traditional mechanistic models [15]. Data-driven models generally outperform mechanistic models in prediction accuracy. For example, in a study by Dammau et al. [25] compared the performance of deterministic and data-driven models in describing fouling in MBRs as measured by the Trans Membrane Pressure (TMP) over 462 days. The data-driven model excelled in scenarios with significant fluctuations, such as low temperatures and varying pH or dissolved oxygen levels.

For generalization, which refers to how well a model performs on unseen or new data, data-driven models generally have better generalization ability, as they can capture complex relationships across varied datasets. For instance, in a study comparing mechanistic and ANNs for predicting tissue-to-plasma concentration ratios in rats, Nestorov et al. found that the ANN demonstrated superior generalizability and accuracy, with a mean prediction error of only 2 % compared to the mechanistic model's 18 % overprediction tendency [26].

In terms of computation time, data-driven models, once trained, can make predictions almost instantly, whereas mechanistic models often require significant computational resources due to the complexity of solving physical equations. In one case study, an LSTM neural network achieved a 30,000-fold computational acceleration compared to a mechanistic PDE model for simulating a synthetic gene circuit [27].

Due to the wide variety of methods and their capabilities within artificial intelligence, different classifications have been suggested. However, the most common way is to categorize these algorithms based on the type of data used for modeling in three broad classes as suggested in Fig. 2a: supervised learning, unsupervised learning, and reinforcement learning [28]. Specifically, supervised learning is a machine learning technique that involves training a model on labeled data, which means that each input is associated with a corresponding output. The goal of supervised learning is to learn a route from input to output that can be used to make predictions on unseen data. Common examples of supervised learning include image classification, speech recognition, and natural language processing [29]. One of the most popular supervised learning algorithms is the decision tree algorithm. Decision Trees (DT) are a visual representation of the decision-making process, where each node represents a decision, and each branch represents a possible outcome. DTs are used for both classification and regression tasks, making them versatile algorithms [30]. The k-means clustering algorithm is one of the most popular unsupervised learning algorithms. K-means clustering is a technique used to partition a dataset into k groups based on similarities between data points. The data points are iteratively assigned by the algorithm to a cluster and the cluster centroid is updated until convergence. K-means clustering is widely used in various

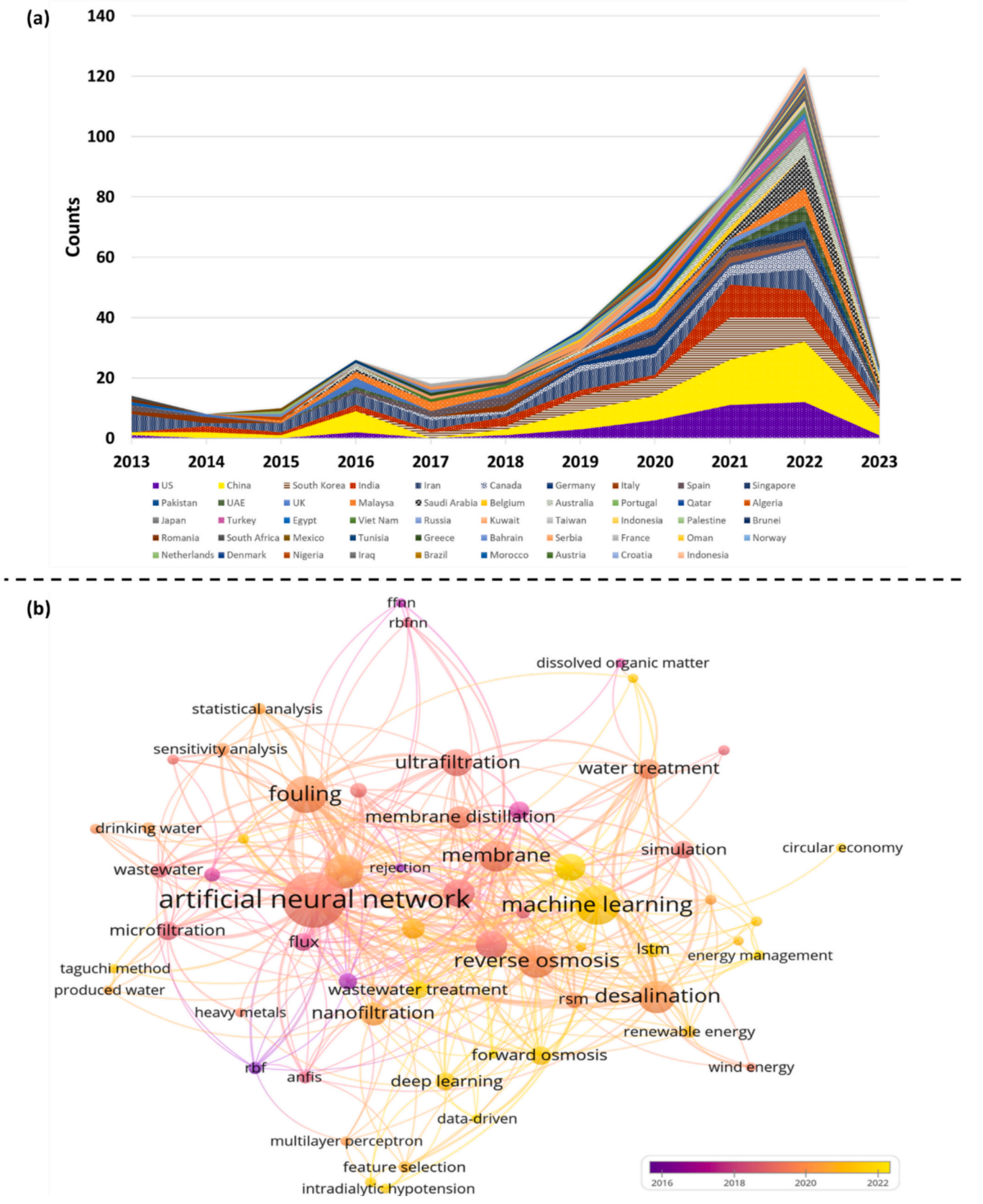


Fig. 1. a) The spatiotemporal trend of articles published in membrane research using artificial intelligence techniques by source country from the year 2013 up to the present time. The legend shows the top 12 contributing countries. b) Co-occurrence network of keywords in papers. The weight of each node determines the number of repetitions of each keyword with a color bar showing how recently it has been used in a publication.

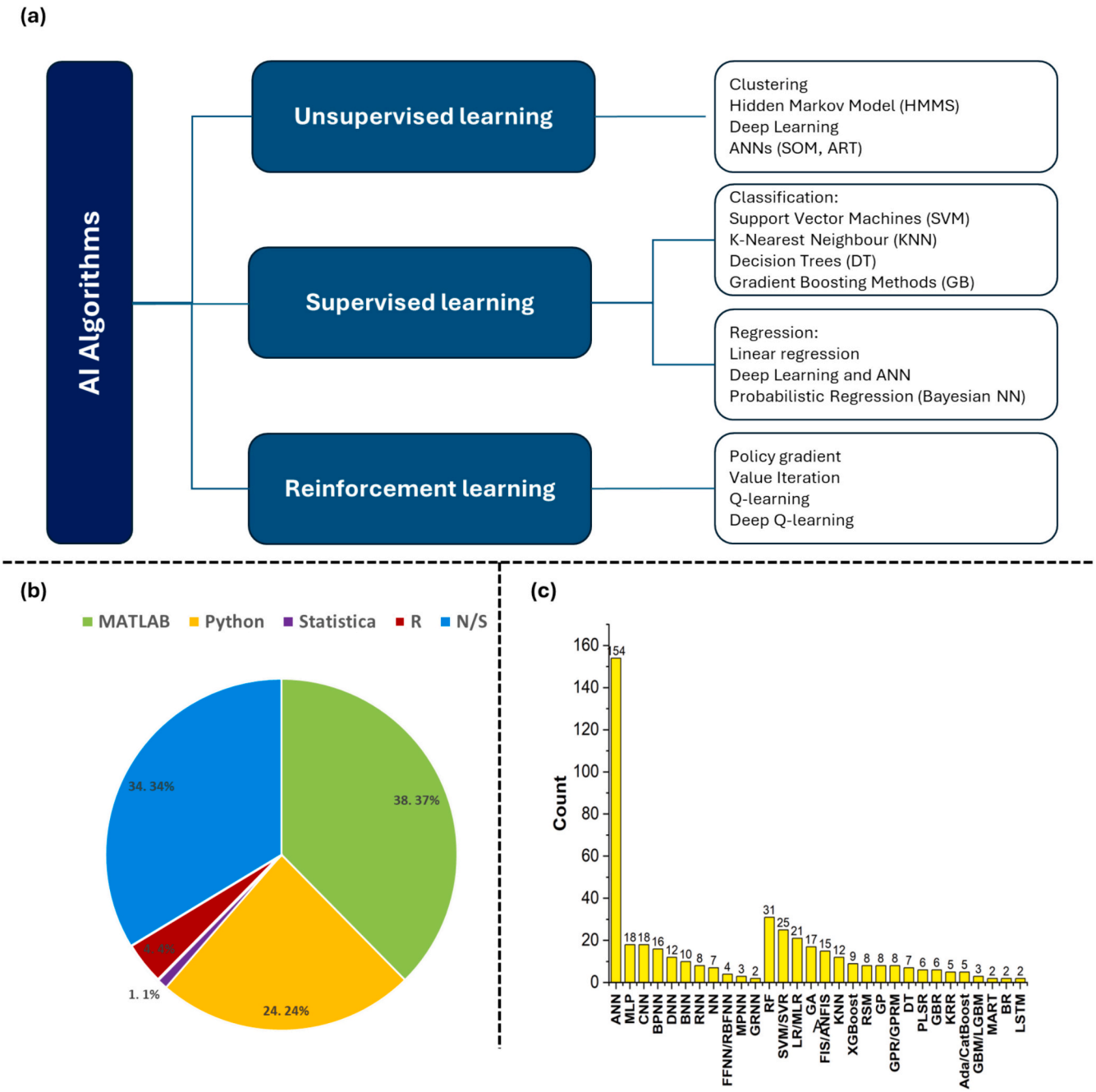


Fig. 2. a) Classification of AI Algorithms Based on the Type of Data Used for Modeling, b) Pie chart Showing the Share of Different Platforms Used for AI Modeling of Membrane Processes c) AI Algorithms Used for Membrane Research by Their Frequency.

applications, including customer segmentation and anomaly detection [31]. Reinforcement learning is a ML technique involving training an agent to interact with an environment aiming at maximizing a reward signal. The agent learns through trial and error by taking actions and observing the resulting rewards. The goal of reinforcement learning is to develop a policy that can make optimal decisions in a given environment [32]. One of the most used reinforcement learning algorithms is Q-learning. Q-learning is a technique used to learn a policy for decision-making in a Markov Decision Process (MDP). The algorithm estimates the optimal action-value function, which is the expected reward for taking a specific action in a given state [33].

In most cases, AI-assisted membrane studies belong to supervised learning in which various methods have been used using different software packages and libraries, such as Python (Scikit-learn) [34], MATLAB [35], R [36]. MATLAB and Python possess an almost equal share in AI-assisted membrane studies with 38 % and 34 % respectively (Fig. 2b). On the other hand, R and Statistica have a much smaller share in studies so far.

MATLAB and Python are favored in membrane studies for their pre-implemented functions and user-friendly syntax, making complex data analysis accessible to researchers without extensive programming backgrounds. They offer strong visualization tools, like MATLAB's built-in plot functions and Python's Matplotlib and Seaborn libraries, aiding in data interpretation. Additionally, their active user communities provide valuable resources, collaboration opportunities, and updates on field advancements. Python's open-source nature and its large, diverse community further enhance its suitability for a wide range of scientific and non-scientific applications. In terms of AI algorithms, so far, over 50

different algorithms have been applied to membrane research. Although, a huge bias can be observed in the mostly used by researchers from Fig. 2c. Among deep learning methods on the left side of Fig. 2c, artificial neural network (ANN) with >150 times of applications, is by far the most preferred algorithm. While, among machine learning methods, random forest (RF) and support vector machine (SVM) are implemented the most. Each AI algorithm has its own logic and procedure for modeling that makes them suitable for specific applications and in the present work, we discuss four major algorithms used in membrane research in the following section.

3.3.1. *Artificial neural network*

An effective and currently popular approach for anticipating and simulating a procedure is through the use of an artificial neural network. ANN is designed to mimic the functionality of the human brain, which involves neurons and weights that are utilized for modeling and predicting various parameters. Additionally, the neurons in an ANN communicate with one another through weights [37].

This type of neural network operates like a black box model, meaning that it is not constrained by physical laws and its parameters may not have any physical significance. The primary aim of ANN is to establish a relationship between independent input variables and dependent output variables [38]. This relationship is derived through a learning process in which the network is fed a dataset, making it a data-driven model. ANN has become increasingly popular due to its ability to quickly and accurately map non-linear relationships between system inputs and outputs, resulting in decreased computational time [39].

The training of an ANN typically involves one of three main learning paradigms: supervised, unsupervised, or hybrid. Multilayer perception (MLP) is one of the most powerful methods of ANN [40]. An MLP, or multi-layer perceptron, is composed of different layers including an input layer, at least one hidden layer, and an output layer as shown in Fig. 3a. The number of neurons or nodes in each layer is determined to match the number of input and output parameters. Depending on the optimal architecture of the ANN model, the hidden layers may have several neurons [39]. During the training phase, the weight of each neuron in the hidden and output layer is computed and assigned. Additionally, a bias or threshold value is included as a weighted unit input for each neuron in the hidden and output layer.

Several MLP-ANN algorithms are developed which Back Propagation Feedforward Neural Network (BPFFNN) is among the most used ones including Radial Basis Function Neural Network (RBFNN) [41], and Deep Neural Network (DNN) [42], Elman Neural Network (ENN), Recurrent Neural Network (RNN) [43].

3.3.2. *Support vector machine*

The Support Vector Machine (SVM) method is a relatively new statistical technique in machine learning that relies on an optimization

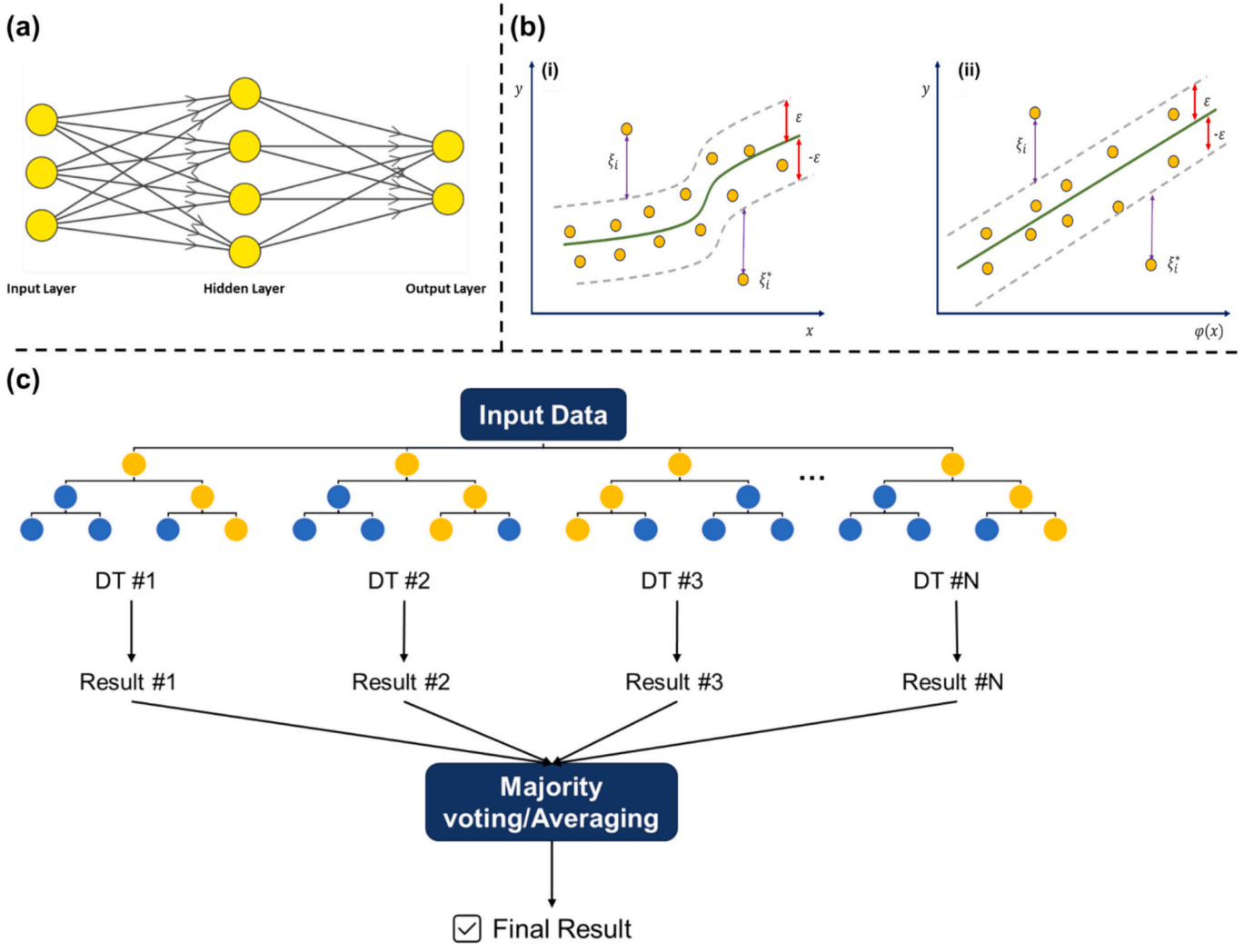


Fig. 3. a) A typical single hidden layer MLP architecture with 4 input, 4 hidden, and 2 output neurons. b) Support vector regression for non-linear response modeling from (i) higher dimensional domain to (ii) lower dimensional domain. c) Random Forest ML model classification procedure and logic.

algorithm developed by Wang et al. [44]. Although originally used for classification tasks, SVM has recently been expanded to address regression problems as well which is called Support Vector Regressor (SVR) [45].

SVM is designed to minimize projected error in the learning model, thereby reducing over-fitting and promoting generalization through the use of structural risk minimization (SRM) [46]. Unlike other methods that pre-determine the structure, SVM allows the training dataset to determine the model's contribution by selecting only the most relevant data samples, which are referred to as 'support vectors' [47]. The modeling procedure, in which data is shifted into a chosen dimensional space is demonstrated in Fig. 3b. The SVM employs an objective function, as shown in Eq. (1).

$$min\frac{1}{2}||\omega||^2 + C\sum_{i=1}^{n}\left(\xi_i + \xi_i^*\right) \quad (1)$$

Given:

$$(\omega\phi(x_i) + b) - y_i \le \varepsilon + \xi_i$$

$$y_i - (\omega\phi(x_i) + b) \le \varepsilon + \xi_i^*$$

Where ω is the direction vector, C represents the adjustment factor: a trade-off between the training error and flatness of the model. Slack variables are shown as ξ_i and ξ_i^*. $\phi(x_i)$ denotes the higher dimensional hyperspace for the input vector x_i. b is the intercept of the regression function and ε is the non-sensitivity coefficient [48].

To prevent the dimensional disaster associated with the high-dimensional quadratic programming of the SVM model, high-dimensional computation is transformed into low-dimensional computation using a kernel function. Various kernel functions are available for SVM modeling, including linear, Gaussian, polynomial, and RBF. [45].

3.3.3. Random forest

Random forest is a model that uses trees and combines the strengths of tree-based models while addressing the limitations of traditional classification or regression trees. Decision tree models are particularly susceptible to overfitting, even with proper hyper-parameter tuning. This means that the decision tree model may not perform well in predicting outcomes on new, unseen data [49]. Random forest, on the other hand, overcomes the overfitting issue by utilizing randomness in selecting bootstrapped datasets and constructing trees based on a random subset of features [50]. This has made the RF model a popular and dependable machine-learning model for predicting the performance of various membrane systems as shown in Fig. 3c.

To train the RF model, there are three primary stages involved: First, the main training dataset is used to create multiple subsets called "bootstrapped datasets". The number of subsets formed is equal to the number of trees in the forest. It is important to note that the size of each bootstrapped dataset is the same as the training dataset, meaning that the same sample can be selected multiple times. Then, each bootstrapped dataset is utilized to generate a tree and a random subset of features in order to construct a tree. Finally, the process in step 2 is repeated to create a new tree with a different bootstrapped dataset and randomly selected features until all the trees in the forest have been formed [51].

As shown in Fig. 3c, random forest uses majority voting or averaging of the predictions of the independent trees to produce an output; reducing the variance and improving the generalization of the model due to the high number of estimators used [52].

3.3.4. Gradient boosting

Gradient boosting methods are a subset of ensemble methods similar to bagging methods (i.e., DT and RF). However, despite the bagging methods, in boosting, the accuracy of prediction is improved through serial learning [53]. Adaptive Boosting (AdaBoost), Categorical Boosting (CatBoost), Gradient Boosting (GB), and Extreme Gradient Boosting (XGBoost) are some of the popular boosting methods used by data scientists and engineers.

In boosting, models are built sequentially by minimizing the errors from previous models while increasing (boosting) the influence of high-performing models [54]. In order to minimize the errors in sequential models, a gradient descent algorithm is added to form the gradient boosting approach. Finally, GB through parallel processing, tree pruning, handling missing values and regularization to avoid overfitting/bias yields XGBoost which is capable of producing superior results using fewer computing resources in the shortest amount of time [55].

The selection of an appropriate AI approach for membrane studies depends on the criteria that are related to the data, type of membrane, and application context. Basic methods, such as decision trees or random forests, should be prioritized for structured or moderately complex data. Sophisticated models, such as neural networks or deep learning, are necessary for more intricate problems. The type of membrane should also be taken into account; for example, polymeric membranes may only require ensemble learning techniques, whereas more complex systems, such as nanocomposite or MMM, typically necessitate advanced models that can encapsulate intricate material interactions. Additionally, the context of the application should be considered. For example, evolutionary algorithms or genetic optimization are advantageous for optimization tasks, such as membrane design, whereas reinforcement learning or time-series models like LSTM are the most effective approaches for real-time monitoring and control, including contamination detection. Hybrid methodologies, which incorporate a variety of AI techniques, are frequently employed to enhance the accuracy and interpretability of forecasts, thereby allowing models to accommodate both the inherent physical limitations and the complexity of the system. It is beneficial to employ and compare multiple proven methods like ANN, RF, SVM, and XGBoost, assessing their performance using accuracy indicators such as R^2, RMSE, MSE, and MAE. These metrics are vital in evaluating AI algorithms in regression tasks.

$$R^2 = \frac{\sum_{i=1}^{n}(P_i - \overline{y_i})^2}{\sum_{i=1}^{n}(y_i - \overline{y_i})^2} \quad (2)$$

$$RMSE = \sqrt{\frac{\sum_{i=1}^{n}(P_i - y_i)^2}{n}} \quad (3)$$

$$MSE = \frac{\sum_{i=1}^{n}(P_i - y_i)^2}{n} \quad (4)$$

$$MAE = \frac{\sum_{i=1}^{n}\left|P_i - y_i\right|}{n} \quad (5)$$

Where n is the number of data points, P_i represents the predicted output of the model, $\overline{y_i}$ denotes the sample data mean, and y_i is the actual output. Ranging from 0 to 1, a higher R^2 value indicates a better-fitted model [56].

Accuracy is the ratio of correct predictions to total predictions made. Precision is the ratio of true positive values to all values that were predicted as positive, while recall refers to the ratio of true positive values to all experimentally positive values [57].

In order to perceive a general idea regarding the accuracy of different AI algorithms applied in membrane research, R^2 can be a suitable indicator considering its non-dimensionality and good availability given it is often reported in relevant studies. Fig. 4, summarizes the prediction accuracy of representative output parameters such as rejection rate, permeate flux, selectivity, and permeability within different membrane filtration process modeling studies.

Furthermore, to validate the robustness of AI simulations, it is crucial

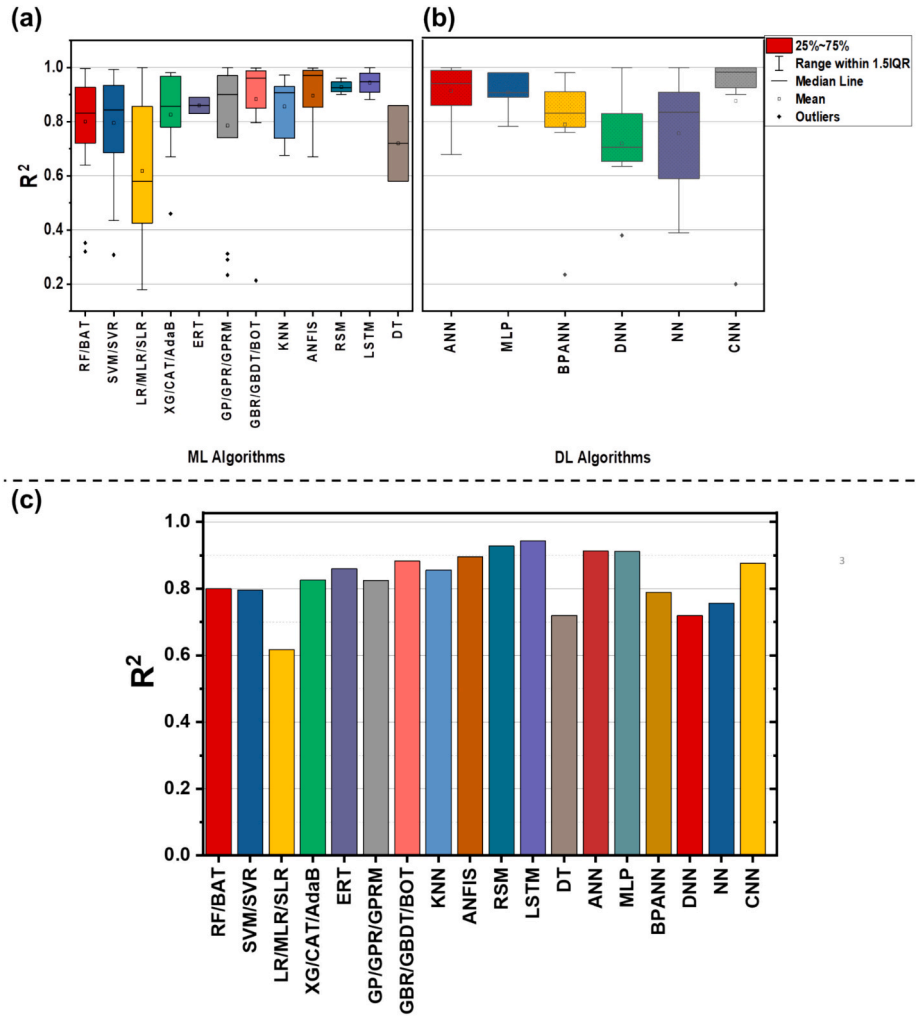


Fig. 4. The prediction accuracy of representative outputs based on the coefficient of determination (R^2) in AI-assisted membrane studies carried out by a) machine learning algorithms and b) deep learning algorithms (cross marker inside each box indicates mean value while the outlier data is shown by dots positioned far from each box). c) Average coefficient of determination for all AI algorithms considered.

to test the model on data that falls outside the distribution of the training set and incorporate domain expert evaluation. Out-of-distribution testing can be achieved by generating synthetic data, collecting diverse real-world samples, and implementing adversarial testing to identify potential vulnerabilities. Simultaneously, involving domain experts in the validation process is essential, particularly for complex simulations. These experts can assess the plausibility of the model's outputs, identify potential biases or inconsistencies, and provide valuable insights into edge cases that may not be captured by standard validation techniques. The combination of out-of-distribution testing and expert evaluation ensures a more comprehensive and reliable validation process, enhancing the credibility and applicability of AI-driven simulations in real-world scenarios.

Very satisfying R^2 value ranges are observed for some machine learning models shown in Fig. 4a, namely gradient boosting algorithms (GBR/GBDT), Gaussian process regression (GP/GPR/GPRM), K-Nearest Neighbors Algorithm (KNN), Adaptive neuro-fuzzy inference system, (ANFIS), random forest (RF), support vector machine/regressor (SVM/SVR) and extreme gradient/Cat/Ada boosting (XG/Car/Adaboost) with an average coefficient of determinations of 0.98, 0.91, 0.91, 0.98, 0.83, 0.86, 0.83 respectively and low variability. To pass judgment on the trustworthiness of other methods such as Response Surface Methodology (RSM), Extremely Randomized Trees (ERT), and LSTM that demonstrate considerably high average R^2 values here, the current number of data points reported in the literature is not enough.

On the other hand, it is perceived from Fig. 4b that DL algorithms – however, less varied – have an overall higher R^2 value compared to ML algorithms, especially ANN, Convolutional Neural Network (CNN), and Multilayer Perceptron (MLP) with average R^2 values of 0.94, 0.99. and 0.91 and a relatively low variation range for values.

Overall, it can be concluded that the selection of the proper method is a crucial stage in artificial intelligence-assisted membrane studies and based on the type of study, data, membrane, and application the accountability of results can vary significantly. Based on the type of study, data, membrane, and application, the accountability of results can vary significantly. It is important to note that the accuracy data presented in this review should be used primarily to gain general insights and comparative ideas about different methods. Readers and researchers should exercise caution when interpreting these results and avoid directly applying them to their specific research contexts without further consideration. A summary of all AI algorithms reported by researchers in membrane studies along with their advantages, disadvantages, type of membranes studies and average accuracy (mean R^2) is summarized in Table 1.

4. AI in membrane technology: Models for purpose

In this section, AI-assisted membrane studies are categorically

Table 1
AI algorithms reported in membrane studies, their advantages, disadvantages, membrane types studied, and average prediction accuracy.

AI algorithm	Advantages	Limitations	Membrane type	Prediction accuracy (average R^2)	Sources
ANN	Flexible, nonlinear, robust, parallelism	Black-box model, big dataset required, intensive computation, prone to local optima	MF, NF, UF, RO, FO, MBR, MD	0.94	([58]; [59]; [51]; [11]; [60]; [61]; [62]; [63]; [64]; [65]; [66]; [67]; [68]; [69]; [70]; [71]; [72]; [73]; [74]; [36]; [75]; [76]; [4]; [77]; [78]; [79])
MLP	Flexible, nonlinear, robust	Big dataset required, intensive computation, prone to overfitting	NF, MBR	0.91	([80]; [37]; [81]; [82]; [83])
BPANN/ BPNN	Flexible, nonlinear, robust, scalable	Big dataset required, intensive computation, prone to overfitting, slow convergence	NF, MBR, UF	0.83	([84]; [6]; [85]; [86])
DNN	High accuracy, feature learning, robust	Large memory requirements, prone to overfitting, Lack of interpretability	NF, UF, MD	0.73	([87]; [42]; [88]; [18]; [89])
NN	Ability to learn and adapt, nonlinear, fault-tolerant,	Big dataset required, intensive computation, prone to overfitting	NF, UF, ED, MD	0.83	([90]; [91]; [92]; [93]; [94]; [95]; [96])
CNN	Effective image and video processor, Hierarchical feature extraction, parameter sharing, translation invariance	Large training data requirements, Limited interpretability, susceptible to adversarial attacks	UF, MD	0.98	([97]; [98]; [42]; [99]; [100]; [101])
RF/BAT	High accuracy, feature importance estimation, robust, parallel processing	Sensitivity to noisy data, not suitable for online learning, black box nature, computationally intensive	NF, UF, RO, FO, MBR, MD	0.84	([102]; [84]; [51]; [80]; [30]; [103]; [81]; [63]; [93]; [6]; [94]; [104]; [72]; [88]; [96]; [52]; [105]; [89])
SVM/SVR	Effective in high-dimensional spaces, good generalization performance, versatile, suitable for both linear and nonlinear problems	Sensitivity to noisy data, not suitable for online learning, black box nature, computationally intensive, prone to overfitting	NF, UF, RO, FO, MBR, MD	0.86	([58]; [84]; [51]; [106]; [80]; [90]; [30]; [107]; [81]; [63]; [108]; [6]; [36]; [76]; [45]; [96]; [89]; [79])
LR/BLR/ MLR/SL	Simple and easy to implement, computationally efficient, can handle multiple input features, can be regularized to avoid overfitting	Assumes a linear relationship between the input features and the output, not suitable for very complex problems, sensitive to irrelevant features or outliers in the database	NF, UF, MBR, MD, ED	0.58	([84]; [90]; [109]; [81]; [64]; [82]; [72]; [36]; [57]; [52]; [79])
XG/CAT/ AdaBoost	High performance, Regularization, Feature importance estimation, robust	Complex, computationally intensive, slow training	NF, RO, MBR, Pervaporation	0.84	([80]; [110]; [111]; [112]; [72]; [83]; [96]; [79])
ERT	High accuracy, feature importance, no prior knowledge needed	Complexity, overfitting, limited interpretability	Gas Separation	0.86	[57]
GP/GPR/ GPRM	Non-parametric, flexible, uncertainty estimates	Computationally intensive, limited scalability, hard to interpret	UF, MBR	0.91	([102]; [84]; [113]; [61]; [114]; [69]; [115]; [116]; [83]; [117])
GBR/GBDT/ BOT	High accuracy, flexible, no prior knowledge needed	Slow training time, overfitting, sensitivity to outliers	MBR, Pervaporation, Gas Separation	0.97	([84]; [80]; [81]; [76]; [118]; [119]; [89]; [79])
KNN	Simple and easy to implement, non-parametric, no training time	Sensitive to the choice of k, slow for large datasets, requires feature scaling	MBR, Gas Separation	0.91	([80]; [107]; [120]; [121]; [36]; [79])
ANFIS	Highly flexible, fast training, transparent, can model complex relationships	Requires expertise, sensitive to initialization, may overfit, not suitable for large datasets	MBR, Gas Separation, RO	0.97	([122]; [61]; [65]; [123]; [73]; [115]; [124]; [74])
RSM	Suitable for complex relationships, efficient optimization, can handle multiple responses	Linearity assumption requires experimental design, sensitive to outliers	NF, FO	0.93	([37]; [65]; [66]; [67]; [87])
LSTM	Suitable for long-term dependencies, high accuracy, learns from past information	Computationally expensive, sensitive to initialization, limited interpretability, risk of overfitting	MBR, UF	0.86	([125]; [126]; [100])
DT	Easy to understand and interpret, can handle both categorical and numerical data	Prone to overfitting, sensitive to small changes in the data, biased towards features with many levels, affected by outliers	Gas Separation, MBR	0.72	([106]; [80]; [30])

analyzed based on the membrane type investigated, the industry sector targeted, and the ultimate objective aimed at.

4.1. AI and microfiltration

Microfiltration (MF) membranes are the oldest of the four pressure-driven membrane processes [127] with separation pressures ranging from 100 to 400 kPa [128], which are the lowest pressure requirements, and highest permeability when compared to the other pressure driven membranes. The membrane separation layer works like a filter, allowing only particles smaller than 100–10,000 nm and molecules with molecular weights <100–500 Da to pass through while blocking larger particles and heavier molecules. [15]. Polymeric materials are most used for MF membranes, however, a wide range of materials are available for application on this membrane type including ceramics where SiO2 and Al2O3 are the most commonly used non-organic MF membrane materials [129].

As shown in Fig. 5a, MF membranes have been spotted in the literature only 14 times which is one of the lowest investigated membrane module types by AI. This can be related to the recent nature of the application of AI in membrane studies as opposed to the long history of MF membranes in the chemical and material engineering field leading to the overlooking of researchers to this particular module. Liu et al. [130] developed an ANN model for the turbulence promoter-assisted Cross-flow Microfiltration (CFMF) process using experimental data, in which the inlet velocity, TMP, and feed concentration were taken as inputs and the Flux Improvement Efficiency (FIE) by turbulence promoter as output with a prediction accuracy estimated by R-value of 0.85. According to the result of the sensitivity analysis based on the correlation coefficient, the filtration time was the most significant input, followed by temperature, cross-flow velocity, feed pH, and TMP. A more recent study by Tanudjaja et al. [131] aimed to improve the prediction of critical flux utilizing machine learning models, and employed RF and NN algorithms on a dataset of 223 data sets from various studies, including nine operational parameters. The results showed that crossflow velocity was the most dominant parameter, followed by foulant size and charge, in determining critical flux via the RF model. Additionally, the NN models provided accurate predictions of critical flux with an R^2 consistently above 0.9. Unlike the previous studies focusing mostly on water/wastewater treatment application of MF membrane, which as shown is the most investigated subject area of AI-assisted membrane studies, Abdullah et al. [132] have implemented the ANN-GA method for tannin reduction and clarification of cashew apple juice using pressure-driven CFMF. ANN has successfully determined the optimum process condition for MF in terms of membrane pore size and TMP in order to maximize the tannin content, turbidity, and color. The study also developed a resistance-in-series model that accurately represented the flux behavior of the microfiltration system.

4.2. AI and ultrafiltration

Ultrafiltration (UF) membranes operate using the same separation principles as MF membranes, with UF driving pressures varying from 100 to 1000 kPa and their pore size ranging from 0.001 to 1 μm [133]. The main difference between the two membrane technologies is their ability to reject smaller molecules like macromolecules, colloids, and solutes with molecular weight larger than 10,000 Da [128]. The simplicity, overall low cost of the process, low energy usage, fewer control techniques, little or less emphasis on process chemicals, mild operating temperature, and high-quality treatment of the UF process are the main reasons why they have replaced conventional purification and disinfection processes [134].

In an effort to thoroughly analyze UF membranes performance and fabrication conditions, four different supervised learning (SL) approaches - multiple additive regression trees (MART), NN, LR, and SVM – have been utilized by Chi et al. [90]. Based on the analysis, the most reliable and trustworthy SL model was identified to be NN with R^2 values of 0.8992 and 0.9559 for the rejection rate and the flux prediction. This model was then used to optimize the fabrication conditions and predict the optimal performance of PVC/PVB ultrafiltration membranes. As can be perceived from Fig. 6, membrane design and fabrication is the most investigated objective of the study by membrane scientists using AI techniques after process optimization.

Similar to single-module processes, AI has shown great potential in simulating and predicting hybrid systems. A recent example is the work of Nam et al. [71] to assess the effectiveness of a FO–UF hybrid system by measuring water flux and RhB removal. To achieve this, the researchers employed RSM and Back Propagation Artificial Neural Network (BPANN) models to forecast and optimize the experimental outcomes using various operational parameters (initial RhB

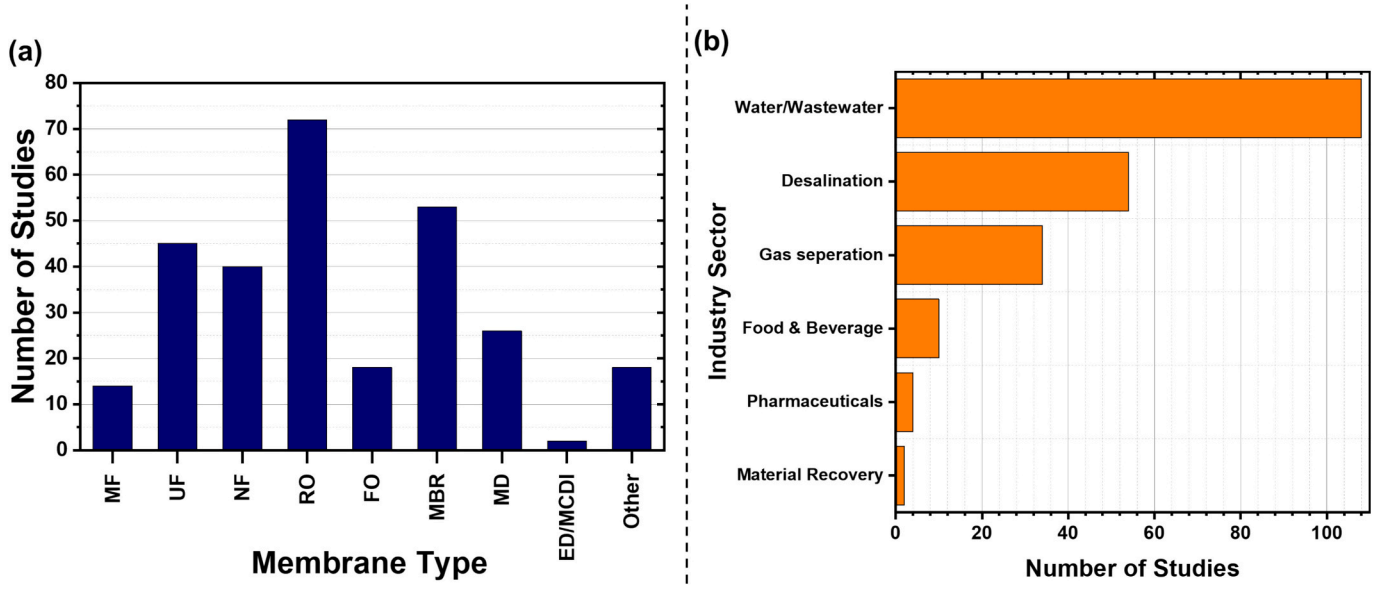


Fig. 5. a) Membrane module type investigated by AI in the literature database analyzed: Microfiltration, Ultrafiltration, Nanofiltration, Reverse Osmosis, Forward Osmosis (FO), Membrane Bioreactor, Membrane Distillation/Pervaporation, Gas Separation Membranes. b) Number of times the different industrial applications have been spotted in the AI-assisted membrane literature analyzed.

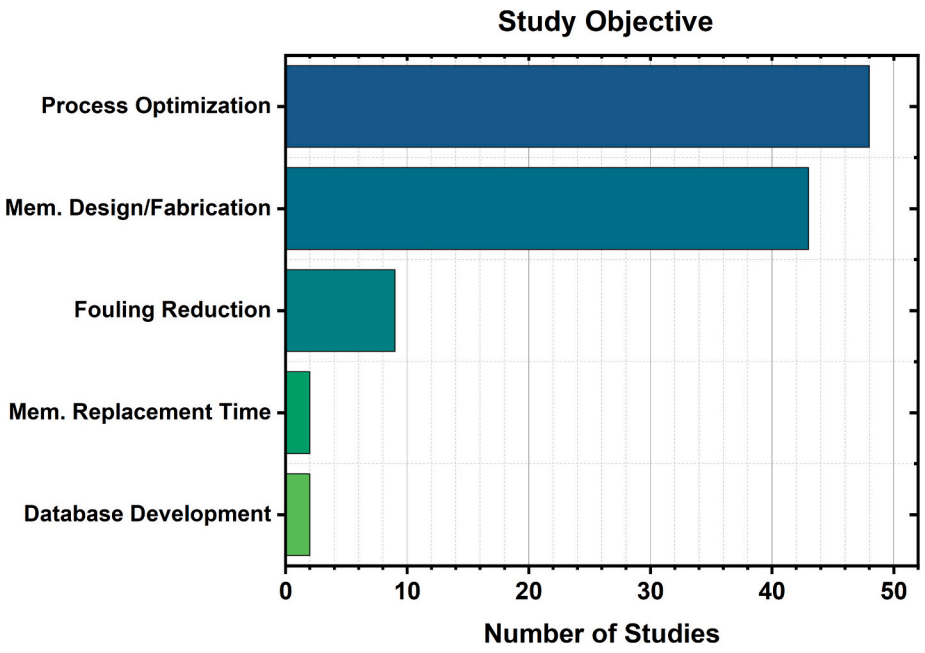


Fig. 6. Most repeated objectives of AI-assisted membrane studies are sorted by their respective number of repetitions in the literature.

concentration, initial pH of feed, and NOM concentration). The results were promising as the quadratic model from the RSM aligned well with the experimental data, and a second-order polynomial equation accurately described the connection between the response and the variable. Ultimately, the analysis identified DS and FS concentrations as the most critical factors influencing water flux and RhB removal.

4.3. AI and nanofiltration

Nanofiltration (NF) membranes are another pressure-driven membrane technology with high rejection efficiencies with most dissolved organic solutes and good salt rejection at salt concentrations varying from 1000 to 2000 ppm [135]. Typical NF pore size ranges from 0.001 to 0.01 μm and MWCO value of 150–2000 Da [136]. The rejection mechanism of nanofiltration membranes is not only based on size exclusion, as seen with other membrane types, but it is a combination of size exclusion, charge repulsion and adsorption [137,138].

An increasing interest can be noticed in the study of organic solvent nanofiltration (OSN) processes with AI in recent years [63,139,140]. A novel quantitative structure–activity relationship (QSAR) is proposed by Ignacz and Szekely [141] for precise prediction of solute rejection in OSN membranes. An extensive database is used for the training of two algorithms of partial least-squares (PLS) regression and message-passing neural networks (MPNN). In the database, molecule structures are stored as SMILES (Simplified Molecular-Input Line-Entry System) which is essentially a direct string representation of a molecule. In this way, a model is able to investigate dipper layers of structural behaviors of membrane, pollutants and solutions used in the process and link the observations with underlying phenomena such as functional groups, and molecular weight. The applied methodology in this article, along with a Graphical User Interface (GUI) rejection prediction tool available at www.osndatabase.com for free usage by researchers, boosts the application of OSN in novel membrane design and different industrial sectors. In another study on NF, Zhang et al. [18] have addressed the data scarcity problem of DL models which generally leads to overfitting, by extracting the natural features of monomer molecular structures and then distorting them in a reasonable manner. The rejection and flux predictions of polyamide NF membranes are carried out by molecular augmentation on DNN with <5 % mean relative error.

Baig et al. [142] have explored the application of machine learning techniques to predict the performance of loose nanofiltration membranes in dye and salt removal from wastewater. Using experimental data from a laboratory setup, four computational models were evaluated: hybrid adaptive neuro-fuzzy inference system (ANFIS), robust-linear regression (RLR), support vector regression (SVR), and multi-linear regression (MLR). The results showed that SVR-M2 performed best for predicting flux and rejection, while ANFIS-M2 excelled in modeling dye/salt rejection. This research demonstrates the potential of combining experimental membrane design with machine learning to optimize the fractionation of divalent salts and dyes in wastewater treatment.

4.3.1. AI and reverse osmosis

Reverse osmosis (RO) membranes are pressure-driven semipermeable membranes with a separation range between 0.1 and 1 nm, used for the removal of solutes such as small molecular weight molecules (e.g., organic compounds) and salts, while allowing water molecules to permeate through the membrane [39]. RO membrane's pore size range is so small that it results in slower permeability, thus requiring the application of high pressures to overcome this limitation [143]. RO membranes can reach the rejection of up to 99.5 % of small particles, making it an appropriate technology for applications requiring high rejection efficiencies such as desalination and treatment, and reuse of wastewater [144].

Operational cost optimization is another important objective that has been at the center of attention for membrane technologies that have found their way into the industry but still need considerable investments compared to conventional technologies [145]. An example of such an endeavor is the work of Nazif et al. [146] in which NN-based optimization of the RO system is practiced in order to reduce fouling, increase membrane life span, and minimize system costs. According to the model's results, controlling inlet pressure, inflow, and system recovery can prolong the membrane's life for small RO systems and minimize cleaning cycles. In another similar study, Srivastava et al. [147] used ANN for early fault prediction and predictive maintenance of RO plants. Five plant parameters namely, pH and TDS of product water, feed and product water flow rates and washing cycles along with plant operation time are used as inputs to predict membrane efficiency with a very high accuracy of R^2 = 99.99 %. However, the results must be approached with caution, since the rather small size of the database, i.e., 75 samples

accompanied by exceptionally high accuracy, suggests the possibility of overfitting.

4.4. AI and forward osmosis

FO membranes are based on osmotic processes that require the use of a higher concentration draw solution (DS) compared to the lower concentration Feed Solution (FS) [148,149]. The high osmotic pressure gradient between the DS and FS drives the water transport mechanism and the net flow of water molecules across the semipermeable membrane [150]. The diluted DS can either be recovered and separated from the water, with an additional separation step, or it can be used as is [151]. In comparison to RO membrane processes, FO does not require any additional significant energy for its operation, however in the case of DS regeneration a significant energy input is needed [152].

Up to now, some critical issues have hindered the widespread industrialization of FO membranes such as reversal solute flux (RSF) and concentration polarization (CP) [153]. Ibrar et al. [111] investigated the ability of several AI algorithms such as ANN, XGBoost, CatBoost, RF and LR in accurate prediction of internal CP with superior performance observed for CatBoost in terms of RSME and R^2 (0.92). This study proves the applicability of AI methods in prediction of very sophisticated physical phenomena laid underneath FO process, especially considering the appropriate feature selection carried out by researchers, ruling out the most correlated parameters (>0.95) in order to obtain a more robust model. On the other hand, there is a challenge to current wastewater treatment technologies due to the presence of trace amount of contaminants known as micropollutants [154]. The process of 97 MPs removal by FO membrane was recently studied by Viet and Jang [155]. It has been discovered that FO has a very low rejection capability for MPs with low molecular weight (2–80 %) with a noticeably precise rejection prediction from the modifier ANFIS method (R^2 = 0.99 and RMSE = 0.56 %).

4.5. AI and membrane bioreactor

Membrane bioreactors (MBRs) are a type of wastewater treatment system that combines both biological treatment processes with membrane filtration and separation technologies [156,157]. A MBR's treatment effectiveness is higher than that of other biological systems due to the extremely dense microbial population that surrounds the membrane surface and ensures that all pollutants are eliminated before the wastewater passes through the membrane [158–160]. In addition, because of the sieving effect of the membranes, pollutants with molecular weights higher than the molecular weight cut-off of the membranes are maintained, bringing them into touch with the microorganisms responsible for their destruction inside the MBR [161].

As mentioned, fouling is a critical issue in membrane studies, as such to emphasize the importance of fouling studies in membranes, it was decided to present the fouling reduction as a separate category in Fig. 6. Fouling plays an even more critical role in the case of membrane bioreactors. Thus, Woo et al. [162] introduce a new technique for estimating the lifespan of a membrane and reducing the expenses associated with membrane maintenance based on monitoring the functional profile of the membrane using functional machine learning. 12 functional parameters related to biological, chemical, and physical properties derived from TMP and chemical dosage are implemented to identify membrane ageing and lifespan through hidden patterns and trends in these functional parameters in a full-scale MBR plant. The findings indicate that the proposed technique improved the efficiency and accuracy of membrane replacement time detection. In another work by Taheri et al. [74], ANFIS and ANN algorithms are implemented to predict TMP in an anaerobic membrane bioreactor-sequencing batch reactor (AnMBR-SBR) during biohydrogen production. The study used organic loading rates, effluent pH, mixed liquor suspended solids, and mixed liquor volatile suspended solids as input parameters. The ANFIS model was trained using hybrid algorithms for TMP prediction, and the Gauss membership function with four membership numbers provided the highest prediction performance. The findings of this study suggest that AI techniques can be a valuable tool in predicting operational parameters in wastewater treatment, during biohydrogen production with R^2 values of at least 0.88.

4.6. AI and membrane distillation

Membrane distillation is a membrane-based separation process that has emerged as a promising technology with broad applications [163]. This process entails the selective vaporization of a liquid across a hydrophobic membrane, driven by a vapor pressure gradient [164]. Unlike conventional separation methods, membrane distillation operates at low temperatures, making it advantageous for separating temperature-sensitive substances. It offers numerous benefits, including high selectivity, energy efficiency, and the ability to handle complex mixtures [165]. Membrane distillation has demonstrated potential in diverse fields such as desalination, wastewater treatment, food processing, and chemical purification [166].

Due to the special characteristics of this membrane module, accurate modeling and estimation of heat transfer is a requirement. Dudchen et al. [91] have presented a method for assessing the accuracy of empirical methods used to approximate membrane permeability and heat transfer rate in MD. They have Neural Networks on process and membrane parameters to improve accuracy, demonstrating that the inclusion of membrane thermal conductivity, pore size, and porosity variables in an empirical MD model substantially improves estimation accuracy. The results suggest that this method can be applied to other membrane separation processes as well. Later, in a hybrid membrane system study, Qamar et al. [42] tested the applicability and prediction accuracy of a NN trained with a UF membrane on membrane distillation. The results showed that the NN was able to accurately estimate biofilm thickness on both types of membranes using OCT images, demonstrating its potential for use in a variety of filtration systems. This technology has the potential to improve water treatment processes by providing real-time monitoring of biofilm growth and its impact on hydrodynamics parameters.

5. Unlocking membrane research potential with AI: Challenges and emerging areas

Up to this point, it has been elaborated that the integration of artificial intelligence into membrane research has the potential to revolutionize the development and optimization of membrane filtration processes for several applications. Process optimization, membrane design and fabrication along with fouling reduction and process control will keep the spotlight on them for future research as the main objectives of implementing AI techniques in membrane research; however, in order to address these matters in a more comprehensive manner it is highly advised to pay attention into the development of extensive, reliable and standard datasets available to all researchers since it can be perceived from Fig. 7a that a very wide range of parameters is being used as input for developing AI models. These inputs fall into 2 main categories membrane characteristics (on the left side of the x-axis) and operational conditions (on the right side of the x-axis). While parameters such as characteristics of stream which essentially includes the concentration of substances in solutions used reported in the format of TOC, TDS, and TSS as well as flow velocity/rate and temperature are repeatedly used as model input it is important to pay attention to the specific requirement of each study regarding membrane type and prediction target for model input selection. For instance, for studies aimed at membrane design and fabrication, membrane characteristics are essential to be considered fully in the dataset whereas, for studies targeting process optimization, operating conditions would suffice. In another example, one should be aware that energy input (heat transfer) and applied voltage/resistance/

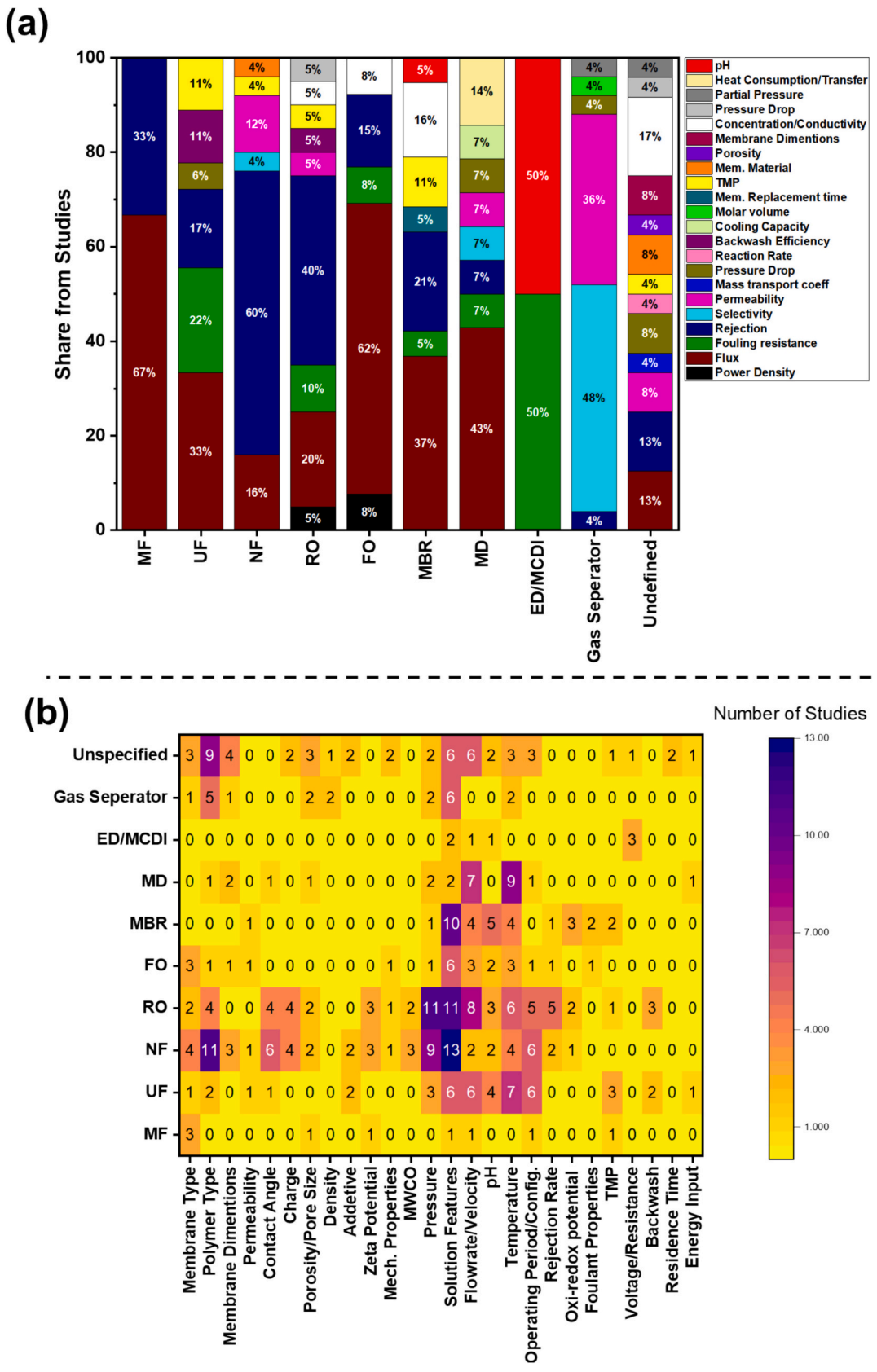


Fig. 7. a) The number of times each input parameter has been identified in the literature for each membrane type. b) Share of each output parameter predicted in literature for each membrane type.

current must be considered for proper modeling of membrane distillation and Electrodialysis (ED) studies, respectively.

As demonstrated in Fig. 1c, emerging research areas such as real-time control systems, integrated membrane and renewable energy resources for better energy management, and micropollutant removal along with developing membrane modules such as forward osmosis and electrodialysis have huge potential to be discovered and investigated in future using novel AI methods. The same goes for AI algorithms and methods. It is observed that more attention has been paid to conducting modeling with DL algorithms such as ANN due to the increase in dataset size in recent years. Furthermore, as RF and SVM are expected to be still the most popular ML methods used, new algorithms based on boosting methods are becoming more and more popular among data scientists mainly due to better performance and less computation time requirement. It is also believed that optimization algorithms coupled with AI models can be the main game changers in future. Algorithms such as Bayesian Optimization [167], Genetic Algorithm (GA) [168], Back Propagation (BP) and Reinforcement learning [169] have shown promising performance in optimizing the decision-making process for AI algorithms. Training as a significantly important step in AI modeling has become more complicated and efficient in recent years with the addition of K-fold cross-validation and grid search [170]; however, this does not necessarily mean that coding has become more difficult, but it is quite the other way around. The introduction of low-code (automated) machine learning libraries such as PyCaret [171] allows the users to import, analyze, train, compare, and showcase their models with the minimum amount of coding and effort.

Additionally, to address the challenge of computational complexity in AI-driven membrane research, two main approaches can be considered:

1) Model Simplification: Developing simpler models that capture essential features while reducing computational demands is a viable strategy. This can involve using dimensionality reduction techniques, feature selection methods, or simplified representations of membrane structures. For instance, treating fragment structures as vertices in graph-based models can significantly reduce complexity, especially when dealing with large polymer systems. Dimensionality reduction is another model simplification approach. Applying techniques such as principal component analysis (PCA) or feature selection to reduce the number of input variables, potentially decreases computational overhead significantly.
2) Efficient Algorithms and Hardware Optimization: Implementing more efficient algorithms, such as advanced machine learning techniques or optimized molecular dynamics simulations, can help mitigate computational complexity. Leveraging high-performance computing resources, including GPUs, and employing parallel processing techniques can also significantly reduce computation time. Furthermore, utilizing techniques like Bayesian optimization can help efficiently navigate complex parameter spaces, potentially reducing the number of iterations required to optimize membrane designs.

Researchers are urged to share their modeling results through user-friendly interfaces. This approach allows others to apply the models, provide feedback, and contribute data, leading to continuous improvement and updates of the original models. This can significantly contribute to the increase in reliability and accuracy of these models and elevates researchers', engineers' and consequently the public's trust leading to faster digitalization of industrial processes.

Model outputs or the prediction targets are categorized based on their share in studies conducted on each membrane type and presented in Fig. 7b. Similar to the model inputs, outputs cover a wide range of parameters with a variable share in each membrane category based on the necessity and application. Nevertheless, flux and rejection dominate in most membrane categories as the main target, and for gas separation, it is mostly selectivity and permeability. With the increasing interest in the development of automated control and diagnostic systems for full-scale membrane setups as well as coupled membranes with other devices, it is expected to see more studies having controlling parameters such as TMP, filter cake thickness, and pH as the target output.

6. Conclusion

In conclusion, this bibliometric review provides valuable insights into the application of artificial intelligence techniques in membrane technologies for separation purposes and its evolvement during the past decade. The integration of AI has significant potential to enhance the performance and sustainability of membrane separation processes for various applications. It is highlighted that AI techniques have been applied successfully in the modeling, optimization, and control of membrane separation processes. Machine learning algorithms such as NN, SVM, DT, and RF have been used to develop predictive models for various types of membranes such as RO, NF, UF, and MF. These models can predict permeate flux, rejection rate, fouling rate, and other performance parameters with high accuracy. Moreover, AI-based optimization algorithms such as GA, PSO, and Bayesian optimization have been used to optimize operating conditions such as pressure, temperature, and flow rate to maximize performance while minimizing energy consumption and fouling. AI-based control strategies such as model predictive control also have been used to maintain optimal operating conditions in real-time and to act as an early fault detection system.

Challenges associated with the integration of AI into membrane research such as data availability issues, model interpretability issues, and computational complexity can be overcome by collaborative and connected research works allowing the growth of reliable databases for different membrane applications, also by developing hybrid models that combine physics-based models with data-driven models (e.g., ANFIS & ANN). Prospects include further development of hybrid models that combine physics-based models with data-driven models to improve model accuracy and interpretability. Moreover, developing explainable AI techniques will enable researchers to understand how the model makes predictions and decisions. This will increase trust in the model's predictions and facilitate its adoption in the industry. AI-assisted membrane studies are promising in emerging fields, including micropollutant removal, FO membranes, membranes in the food and beverage industry, circular economy, and energy management.

CRediT authorship contribution statement

Mehryar Jafari: Writing – original draft, Visualization, Investigation, Formal analysis, Data curation. **Christina Tzirtzipi:** Methodology, Investigation, Data curation. **Bernardo Castro-Dominguez:** Writing – review & editing, Supervision, Project administration, Funding acquisition.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Mehryar Jafari reports financial support was provided by Engineering and Physical Sciences Research Council. If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.jwpe.2024.106532.

Data availability

No data was used for the research described in the article.

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